Assignment 3: congestion control comparison

Completed matched destinations: 3/3. Algorithms: cubic, reno, bbr, a3_cc. Repetitions per algorithm: 3. Duration: 10 s. Sampling: 0.2 s. Seed: 422. Host kernel: 6.18.33.2-microsoft-standard-WSL2.

Test order: seeded initial shuffle; rotate by repetition and candidate index. Exact order, settings, and failures are saved in manifest.json and experiment.log. Only complete groups with matching settings enter comparisons. Successful partial runs remain in per-test-statistics.csv.

Per-run goodput uses acknowledged-byte differences divided by actual elapsed time. Throughput and RTT aggregates are the mean and sample standard deviation across runs. Per-run RTT is the mean sampled smoothed RTT. Retransmitted segments are a loss proxy, not an exact packet-loss probability.

dal.speedtest.clouvider.net:5200 | reno | 3 runs: goodput 253.395 ± 7.888 Mbit/s; RTT 79.114 ± 2.312 ms.

dal.speedtest.clouvider.net:5200 | cubic | 3 runs: goodput 133.716 ± 119.746 Mbit/s; RTT 47.540 ± 22.264 ms.

dal.speedtest.clouvider.net:5200 | bbr | 3 runs: goodput 260.518 ± 5.677 Mbit/s; RTT 75.082 ± 1.320 ms.

dal.speedtest.clouvider.net:5200 | a3_cc | 3 runs: goodput 60.438 ± 28.944 Mbit/s; RTT 34.010 ± 1.093 ms.

185.152.66.67:5201 | cubic | 3 runs: goodput 259.124 ± 8.588 Mbit/s; RTT 62.337 ± 12.217 ms.

185.152.66.67:5201 | bbr | 3 runs: goodput 249.258 ± 8.464 Mbit/s; RTT 75.228 ± 5.726 ms.

185.152.66.67:5201 | a3_cc | 3 runs: goodput 9.690 ± 13.706 Mbit/s; RTT 33.395 ± 3.030 ms.

185.152.66.67:5201 | reno | 3 runs: goodput 211.402 ± 51.549 Mbit/s; RTT 50.627 ± 22.516 ms.

96.45.41.167:30001 | a3_cc | 3 runs: goodput 117.212 ± 61.154 Mbit/s; RTT 121.351 ± 4.099 ms.

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96.45.41.167:30001 | reno | 3 runs: goodput 161.616 ± 6.207 Mbit/s; RTT 123.553 ± 2.781 ms.

96.45.41.167:30001 | cubic | 3 runs: goodput 164.398 ± 4.176 Mbit/s; RTT 121.703 ± 3.684 ms.

96.45.41.167:30001 | bbr | 3 runs: goodput 156.326 ± 18.618 Mbit/s; RTT 125.369 ± 13.224 ms.

Limitations: sequential public-server tests cannot hold routes, cross-traffic, or server limits constant. Small differences relative to run variability do not establish an advantage. These single-flow tests do not establish fairness or convergence.

AI assistance: we used AI tools to generate and revise the majority of the controller, loader, Python experiment code, tests, and plotting/reporting code, and to help refine the design, analysis, and explanations.

Per-run statistics; * = excluded destination group

Destination	Algorithm	Run	Goodput (Mbit/s)	RTT (ms)	Retransmits
dal.speedtest.clouvider.net:5200	reno	1	260.738	77.639	19
dal.speedtest.clouvider.net:5200	cubic	1	269.184	73.249	5
dal.speedtest.clouvider.net:5200	bbr	1	265.854	75.601	1
dal.speedtest.clouvider.net:5200	a3_cc	1	87.525	33.262	0
dal.speedtest.clouvider.net:5200	cubic	2	41.994	34.697	6
dal.speedtest.clouvider.net:5200	bbr	2	254.553	76.063	1
dal.speedtest.clouvider.net:5200	a3_cc	2	29.939	35.265	1
dal.speedtest.clouvider.net:5200	reno	2	245.057	81.779	51
dal.speedtest.clouvider.net:5200	bbr	3	261.146	73.581	3
dal.speedtest.clouvider.net:5200	a3_cc	3	63.850	33.504	0
dal.speedtest.clouvider.net:5200	reno	3	254.388	77.924	37
dal.speedtest.clouvider.net:5200	cubic	3	89.970	34.675	4
185.152.66.67:5201	cubic	1	249.887	59.624	81
185.152.66.67:5201	bbr	1	243.549	80.651	0
185.152.66.67:5201	a3_cc	1	0.957	36.709	5
185.152.66.67:5201	reno	1	270.919	76.622	3
185.152.66.67:5201	bbr	2	258.982	75.791	161
185.152.66.67:5201	a3_cc	2	25.487	30.765	0

Per-run statistics; * = excluded destination group

Destination	Algorithm	Run	Goodput (Mbit/s)	RTT (ms)	Retransmits
185.152.66.67:5201	reno	2	182.461	38.026	701
185.152.66.67:5201	cubic	2	260.616	51.705	298
185.152.66.67:5201	a3_cc	3	2.627	32.710	6
185.152.66.67:5201	reno	3	180.827	37.232	450
185.152.66.67:5201	cubic	3	266.868	75.683	1
185.152.66.67:5201	bbr	3	245.242	69.241	1439
speedtest.wdc2.us.leaseweb.net:5201*	bbr	1	247.040	83.998	7
speedtest.wdc2.us.leaseweb.net:5201*	reno	2	272.102	72.915	0
speedtest.wdc2.us.leaseweb.net:5201*	cubic	2	267.731	70.988	60
speedtest.wdc2.us.leaseweb.net:5201*	bbr	2	265.825	75.025	1
speedtest.wdc2.us.leaseweb.net:5201*	a3_cc	2	50.674	30.172	0
speedtest.wdc2.us.leaseweb.net:5201*	cubic	3	267.641	74.453	2
speedtest.wdc2.us.leaseweb.net:5201*	bbr	3	259.486	80.337	3
speedtest.wdc2.us.leaseweb.net:5201*	a3_cc	3	37.328	30.733	0
speedtest.wdc2.us.leaseweb.net:5201*	reno	3	262.774	74.276	6
nl.speed.vps1.net:5201*	bbr	1	178.251	111.774	7
nl.speed.vps1.net:5201*	a3_cc	1	21.221	109.595	0
nl.speed.vps1.net:5201*	cubic	2	173.944	113.918	1

Per-run statistics; * = excluded destination group

Destination	Algorithm	Run	Goodput (Mbit/s)	RTT (ms)	Retransmits
nl.speed.vps1.net:5201*	bbr	2	175.549	113.279	2
nl.speed.vps1.net:5201*	a3_cc	2	11.442	108.681	0
nl.speed.vps1.net:5201*	reno	2	174.541	116.610	1
nl.speed.vps1.net:5201*	bbr	3	178.626	110.023	4
nl.speed.vps1.net:5201*	a3_cc	3	164.047	115.386	1
nl.speed.vps1.net:5201*	reno	3	178.526	112.005	1
nl.speed.vps1.net:5201*	cubic	3	176.705	112.064	118
speed.netfiber.net.il:5201*	cubic	1	108.692	174.591	0
speed.netfiber.net.il:5201*	bbr	1	99.508	181.959	0
speed.netfiber.net.il:5201*	a3_cc	1	100.915	178.043	27
speed.netfiber.net.il:5201*	reno	1	110.467	174.067	26
speed.netfiber.net.il:5201*	bbr	2	106.634	174.868	0
speed.netfiber.net.il:5201*	a3_cc	2	95.136	183.325	9
speed.netfiber.net.il:5201*	reno	2	110.993	172.798	120
speed.netfiber.net.il:5201*	cubic	2	108.816	175.555	1
speed.netfiber.net.il:5201*	a3_cc	3	99.134	174.832	66
speed.netfiber.net.il:5201*	bbr	3	102.841	179.293	6
iperf.pendc.com:5201*	a3_cc	1	9.959	167.351	1

Per-run statistics; * = excluded destination group

Destination	Algorithm	Run	Goodput (Mbit/s)	RTT (ms)	Retransmits
iperf.pendc.com:5201*	reno	1	105.495	180.607	0
iperf.pendc.com:5201*	a3_cc	2	2.714	169.951	0
iperf.pendc.com:5201*	reno	2	118.065	165.310	0
iperf.pendc.com:5201*	cubic	2	71.822	165.529	0
iperf.pendc.com:5201*	bbr	2	107.679	173.480	0
iperf.pendc.com:5201*	reno	3	110.512	171.497	123
iperf.pendc.com:5201*	cubic	3	20.153	164.490	0
iperf.pendc.com:5201*	bbr	3	111.351	169.192	0
iperf.pendc.com:5201*	a3_cc	3	101.183	171.211	0
96.45.41.167:30001	a3_cc	1	46.615	116.674	0
96.45.41.167:30001	reno	1	165.653	121.293	13
96.45.41.167:30001	cubic	1	169.220	117.690	7
96.45.41.167:30001	bbr	1	167.845	116.564	0
96.45.41.167:30001	reno	2	154.468	126.659	0
96.45.41.167:30001	cubic	2	161.950	122.486	0
96.45.41.167:30001	bbr	2	166.287	118.968	1
96.45.41.167:30001	a3_cc	2	153.880	123.063	8
96.45.41.167:30001	cubic	3	162.023	124.933	19

Per-run statistics; * = excluded destination group

Destination	Algorithm	Run	Goodput (Mbit/s)	RTT (ms)	Retransmits
96.45.41.167:30001	bbr	3	134.847	140.576	0
96.45.41.167:30001	a3_cc	3	151.141	124.315	5
96.45.41.167:30001	reno	3	164.727	122.706	0

dal.speedtest.clouvider.net:5200 | matched settings; individual repetitions

cubic

reno

bbr

a3_cc

Run 1
Run 2
Run 3

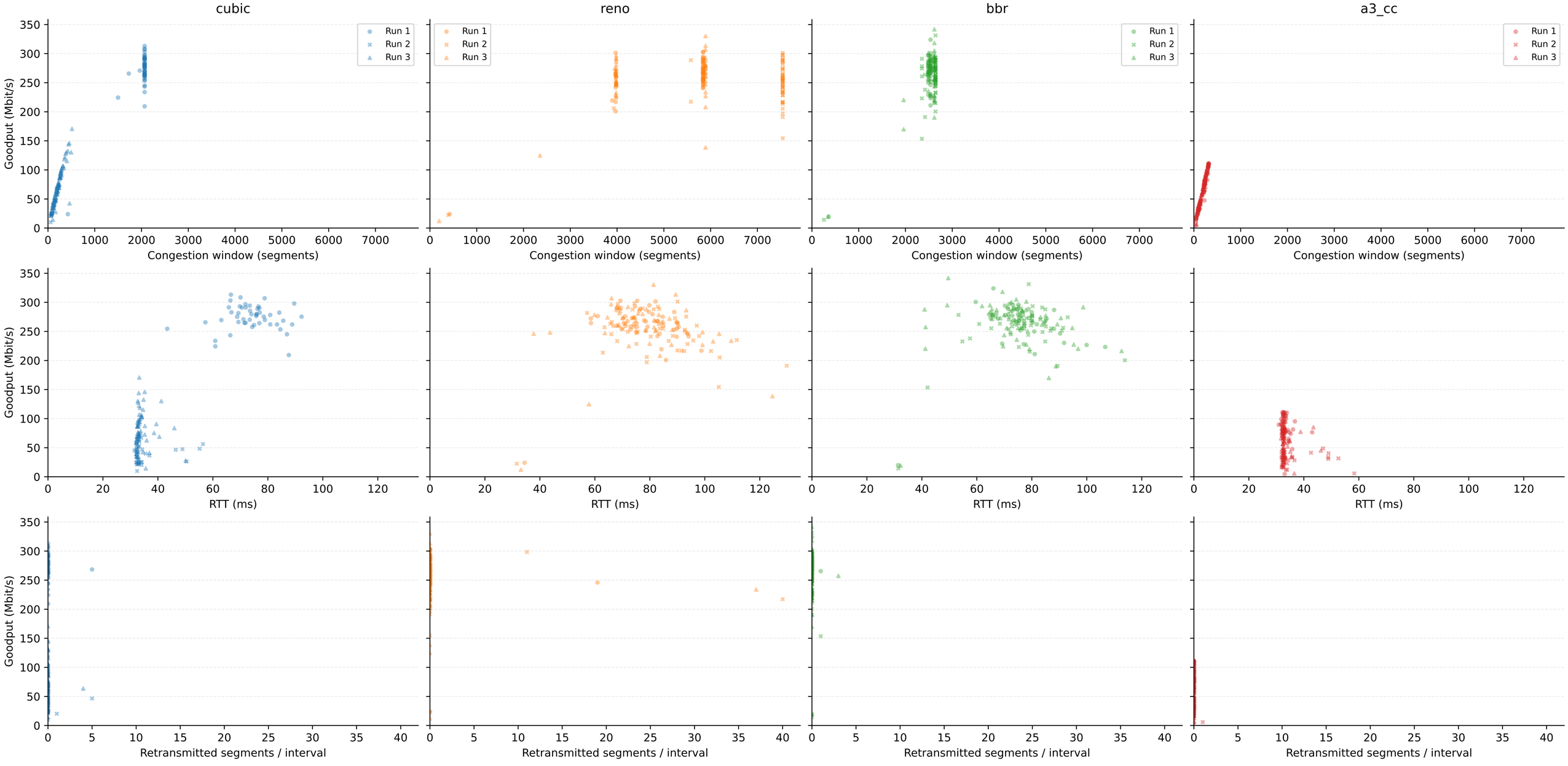
Congestion window (segments)

RTT (ms)

Retransmitted segments / interval

Goodput (Mbit/s)

Time (s)


dal.speedtest.clouvider.net:5200 | all repetitions; common scales per metric
cubic
reno
bbr
a3_cc
Run 1
Run 2
Run 3
Goodput (Mbit/s)
Congestion window (segments)
RTT (ms)
Retransmitted segments / interval